

Lab 4 – Frequency Domain Operations

This lab deals with digital image enhancement. Unlike in the last lab, where we performed spatial domain processing, here we will work in the frequency domain. In the frequency domain, convolution in the spatial domain becomes multiplication, which allows for efficient filtering and precise control of filter properties.

Learning Objectives

- You know how to filter an image in frequency domain.
- You know how to implement a Wiener filter to remove motion blur.

1 Filtering in the Frequency Domain

In the following we want to apply a simple averaging filter h (with varying filter size) to a grayscale image (`MenInDesert.jpg`). This image is shown in Figure 1.

1. Load and display the image (`MenInDesert.jpg`). What are the image dimensions?
2. Compute the image's two-dimensional Fourier transform F . One way to do this is employing `scipy.fft.fft2()`. Which data type does the transformed image have? Read the corresponding documentation, in particular the part about zero-padding.
3. Display the original image and the magnitude (absolute value) of its Fourier transform. Use both a linear and a log-scale.
4. Transform the image back into real space using the `scipy.fft.ifft2()`-command. Compare the data types of the original image and the back-transformed image. How can the original data type be obtained?
5. Create and apply a simple 9×9 averaging filter and apply it to the original image using the `convolve2d()`-command. This carries out the convolution in real space.



Figure 1: Men in desert—the reference image

6. Now, compute the filter spectrum H . Apply it to the transformed image by multiplying both spectra ($G = F \cdot H$) and convert the product back into real space.
7. Make sure that both computations (in real and in frequency space) feature the same result.
8. Compare the computation times for both approaches. Then, increase the size of both image and filter mask by at least a factor of 2 and carry out the same comparison. How does this impact runtime?

2 Image Reconstruction Using a Wiener Filter

In this exercise we intend to reconstruct the blurred image $g(x, y)$, shown in Figure 2, by means of a 2D Wiener filter. The reconstructed image is denoted as f . In the frequency domain, the Wiener filter can be written as

$$\hat{F}(u, v) = \frac{H^*(u, v)}{|H(u, v)|^2 + K} G(u, v)$$

Here, the asterisk denotes the complex conjugate and the parameter K is a real-valued constant.

1. The blurred image was computed from the original image by applying a motion blur filter h , see `exercise2Lab4_motion_blur_generator.py`. It contains two parameters: the direction of the motion and the shift.



Figure 2: Blurred image with hidden text

2. Try to reconstruct the image (below with the blurred text) by using estimates for the direction of motion and the shift. Which parameter value K yields the best results? What does the original text say?